UAV Dataset Summary Report

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Experimental Flight Testing of an Adaptive Autopilot with Parameter Drift Mitigation Yin Yong Chee, Parham Oveissi, Siyuan Shao, Joonghyun Lee, Juan A. Paredes, Dennis S. Bernstein, Ankit Goel Abstract? This paper modi?es an adaptive multicopter au- topilot to mitigate instabilities caused by adaptive parameter drift and presents simulation and experimental results to vali- date the modi?ed autopilot. The modi?ed adaptive controller is obtained by including a static nonlinearity in the adaptive loop, updated by the retrospective cost adaptive control algorithm. It is shown in simulation and ph...

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Unmanned Aerial Vehicle (UAV) path planning in unknown environments continues to pose a significant challenge, as Deep Reinforcement Learning (DRL) solutions are often severely hampered by slow convergence rates as well as unstable training dynamics. To address this gap, we introduce a Genetic Programming?seeded Soft Actor? Critic (GP+SAC) approach in which Genetic Programming produces high?quality trajectories that are introduced into the replay buffer of SAC as a ?warm?start? policy to prevent wasteful early exploration. Through experiments in three benchmark grid environments, we demonstrate that GP+SAC converges significantly more rapidly than the FA?DQN baseline, achieving superior returns in fewer episodes while capitalizing on the same reward design. We show that in large environments, GP+SAC achieved a mean path length of 30.55 units as compared to FA?DQN?s 28.38, thus validating that rapid convergence has no tradeoff in path efficiency. Observably, results also show that as much as GP+SAC obtains superior cumulative rewards, there is a visible fluctuation in the level of training that is indicative of instabilities under very constrained environments. Numerical evaluations...

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https://doi.org/10.31449/inf.v49i2.7178 Informatica 49 (2025) 249?256 249 Optimizing UAV Trajectories with Multi-Layer Artificial Neural Networks Talib Ahmad Almseidein1*, Ala Alzidaneen2 1Department of Basic and Applied Science, Shoubak University College, Al-Balqa Applied University, Al-Salt 19117, Jordan. Talib_m@bau.edu.jo. 2Karak University Collage, Al-Balqa Applied University, Jordan. zidaneenala@bau.edu.jo *Corresponding author Keywords: UAVs, artificial neural networks, optimization, trajectory prediction Received: May 19, 2024 As Unmanned Aerial Vehicles (UAVs) are considered an essen...

File: GJETA-2025-0217.pdf

This study presents a generative design -based optimization of a hexacopter drone frame intended for agricultural spraying applications. Utilizing artificial intelligence -driven algorithms within Autodesk Fusion 360, the research explores multiple frame con figurations based on structural constraints and performance objectives. Finite Element Method (FEM) simulations, conducted through both ANSYS and Fusion 360, modeled dynamic operational scenarios such as landing impacts and aerial collisions. These simulat ions enabled the identification of overstressed regions, leading to iterative material removal and geometric refinement. The finalized frame design achieved a mass of 2.784 kg, translating to a 7.2% to 38.1% reduction compared to traditional 10-liter agricultural drone frames, while maintaining a high safety factor range of 9.60 to 15 and a maximum displacement of just 1.288 mm. Furthermore, flight performance analysis indicated a linear reduction in throttle requirement from 80% to 32.6% as tank weight dec reased from 16.88 kg to 6.88 kg, thereby extending flight duration from 6 to 14.6 minutes. These results demonstrate the effectiveness of generative design for developing ...

File: NOMA-Enabled_UAV_Trajectory_Optimization_Using_Deep_Reinforcement_Learning_for_Efficient_IoT

Non-orthogonal multiple access (NOMA) is a promising technology for future networks because it can allow many sensors to connect and share the same resources at the same time. This provides a spectrally efficient method in situations where spectrum is limited but number of sensors are enormous. In this paper, we extend NOMA to unmanned aerial vehicles (UA Vs) to solve their limited flight time issue while collecting data from sensors that are distributed randomly across an open field. The objective of this study is to

design an efficient UA V trajectory planning scheme that maximizes the total collected sensor data within the available flight time. To achieve this, we model the UA V trajectory optimization as a Markov Decision Process and employ a Deep Q-Network (DQN) to adaptively select UA V movements. A novel reward function is designed to simultaneously guide the UA V toward sensor locations, avoid unnecessary detours, and enhance data collection efficiency under uplink NOMA. Simulation results demonstrate that the proposed NOMA-enabled DQN framework achieves up to 93.5% average data collection, offering an improvement of approximately 19% compared to the traditional TDMA-based...

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Journal of Physics: Conference Series PAPER ? OPEN ACCESS Learning-Based UAV Coverage-Aware Path Planning in Large-scale Urban Environments To cite this article: Zhiyi Shi et al 2023 J. Phys.: Conf. Ser. 2506 012005 View the article online for updates and enhancements. You may also like Helical morphology-inspired bistable gripper for UAV upward perching and grasping in field environment Xuanchun Yin, Shengjian Wen, Junliang Xie et al. - Virtual-target-based reactive and non- cooperative obstacle avoidance: application in low-altitude autonomous aerial navigation in outdoor unstructured enviro...

File: Trajectory-Optimization-Strategy-for-Networked-UAVs-Based-on-Deep-Learning.pdf

The rapid advancement of networked UAV technology has broadened its applications in logistics, disaster relief, agriculture, and military. However, precise trajectory planning in complex dynamic environments remains a key challenge. Traditional methods suffer from low computational efficiency and poor real-time response, limiting practical use. This paper proposes a deep learning-based trajectory optimization strategy for networked UAVs, integrating reinforcement learning and deep neural networks to autonomously learn environmental information and collaborative patterns. Simulation experiments in realistic scenarios validate its effectiveness: the method enhances trajectory planning accuracy and robustness while reducing computational complexity and improving real-time response. This research provides new theoretical and technical foundations for promoting networked UAV applications. KEYWORDS Deep Learning, Networked UA Vs, Trajectory Optimization, Strategy INTRODUCTION In recent years, the vigorous development of unmanned aerial vehicle (UAV) technology has brought broad application prospects (Fei et al., 2023; Rinaldi & Primatesta, 2024). UAVs have become indispensable tools in m...

File: agri_spray_framework.pdf

As the world?s population is expected to increase, so is the global demand for food. Sustainable intensification via precision agriculture of existing farms can increase crop production. Agricultural spray drones have recently taken a physical role within precision agriculture, such as aerial application of fluids, solids, and biological control agents but have difficulties spraying in uncontrolled environments caused by wind shifting spray material away from intended target areas. This work proposes an efficient physics-based framework to provide drone operators with trajectory and spray nozzle configuration for optimal target crop-dusting to mitigate spray drifts while providing quantitative approximations of spray particle trajectory and ground concentration. The framework is coupled with a machine-learning algorithm (MLA) to aid users in their search for optimal results and includes two decoupled models that simulate wind and spray particle trajectories. In the model problem, a genetic algorithm (GA) is used to optimize the system where the optimal trajectory and spray nozzle configuration resulted in 64% of crop targets hit while only losing minimal spray material from spray d...

File: agricultural_hexacopter_design.pdf

Farming technologies have considerably evolved over the last decades considering new agricultural problems and constraints. This paper discusses a Moroccan project based on autonomous spraying drones developed during the last three years in collaboration between Moroccan Industry Services & Engineering company, the INRA institute and the ONCA office. The main purposes of this project is to assist Moroccan farmers improve their performance and enhance their efficiency using chemicals to withstand different diseases, increase crop yields (crops and orchards) and monitor their growth. Also, using our drones helped farmers during the last two years protect themselves and the environment from chemicals and reduce the use of water. Many operations and tests have been carried out in cooperation with the two departments (INRA and ONCA) to highlight the efficiency of our system and to develop new processes together considering this new technology. This paper gives an overview of different experiences conducted, the main results obtained and the technical descriptions of our hexacopter.

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Zero Dynamics Stability of a Multirotor UA V with a Heavy Payload Sander Doodeman 1, Paula Chanfreut 1, Elena Torta 1, Duarte Antunes 1 Abstract? This paper studies the zero dynamics stability of a multirotor Unmanned Aerial Vehicle (UA V) carrying a rigidly attached heavy payload. In particular, we focus on controlling the position of a virtual task-relevant point of interest (POI) on the UA V and investigate how its stability depends on payload placement. Taking the POI position and the yaw angle as output, we prove that the zero dynamics are Lyapunov stable only if this POI is above the sys...

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UA V-VLRR: Vision-Language Informed NMPC for Rapid Response in UA V Search and Rescue Yasheerah Yaqoot?, Muhammad Ahsan Mustafa?, Oleg Sautenkov, Artem Lykov, Valerii Serpiva, and Dzmitry Tsetserukou Abstract? Emergency search and rescue (SAR) operations often require rapid and precise target identification in com- plex environments where traditional manual drone control is inefficient. In order to address these scenarios, a rapid SAR system, UA V-VLRR (Vision-Language-Rapid-Response), is developed in this research. This system consists of two aspects: 1) A multimodal system which harnesses th...

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1 Flight Control for UA V Loitering Over a Ground Target with Unknown Maneuver Fei Dong, Keyou You, Senior Member, IEEE, Jiaqi Zhang Abstract?This paper proposes a ?ight controller for an un- manned aerial vehicle (UA V) to loiter over a ground moving target (GMT). We are concerned with the scenario that the stochastically time-varying maneuver of the GMT is unknown to the UA V , which renders it challenging to estimate the GMT?s motion state. Assuming that the state of the GMT is available, we ?rst design a discrete-time Lyapunov vector ?eld for the loitering guidance and then design a discre...

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1 FALCON-S: Fixed-wing ground-effect Aerodynamics Simulator and Flight Control Learning Suite Matteo El Hariry 1 Pedro Lima 1 Andrej Orsula 1 Matthieu Geist 2 Miguel Olivares-Mendez1, Abstract?We present a modular, high-fidelity simulation framework for the development and benchmarking of flight control strategies in fixed-wing aerial robots operating near the ground. Unlike existing simulators that rely on simplified or hover-oriented dynamics, our framework models full 6DoF rigid-body physics, semi-empirical ground-effect aerodynamics, actuator dynamics, sensor noise, and environmental distu...

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This research contributes to the security design of an advanced smart drone swarm network based on a variant of the Blockchain Governance Game (BGG), which is the theoretical game model to predict the moments of security actions before attacks, and the Strategic Alliance for Blockchain Governance Game (SABGG), which is one of the BGG variants which has been adapted to construct the best strategies to take preliminary actions based on strategic alliance for protecting smart drones in a blockchain-based swarm network. Smart drones are artificial intelligence (AI)-enabled drones which are capable of being operated autonomously without having any command center. Analytically tractable solutions from the SABGG allow us to estimate the moments of taking preliminary actions by delivering the optimal accountability of drones for preventing attacks. This advanced secured swarm network within AI-enabled drones is designed by adapting the SABGG model. This research helps users to develop a new network-architecture-level security of a smart drone swarm which is based on a decentralized network. Keywords: Drone, swarm, Blockchain Governance Game; artificial inelligence, mixed game; stochastic ...

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arXiv:2006.01371v2 [eess.SP] 5 Nov 2021 Energy-Ef?cient Cyclical Trajectory Design for UA V -Aided Maritime Data Collection in Wind Yifan Zhang, Jiangbin Lyu, Member , IEEE, and Liqun Fu, Senior Member , IEEE Abstract?Unmanned aerial vehicles (UA Vs), especially ?xed- wing ones that withstand strong winds, have great potential for oceanic exploration and research. This paper studies a UA V-aided maritime data collection system with a ?xed-wing UA V dispatched to collect data from marine buoys. We aim to minimize the UA V?s energy consumption in completing the tas k by jointly optimizing the co...

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The performance optimization of UA V communication systems requires the joint design of UA V trajectory and communication ef?ciently. To tackle the challenge of in?nite design variables arising from the continuous- time UA V trajectory optimization, a commonly adopted approach in the existing literature is by approximating the UA V trajectory with piecewise-linear path segments connected via a ?nite number of waypoints in three- dimensional (3D) space. However, this approach may still incur prohibitive computational complexity in practice when the UA V ?ight period/distance becomes long, as the distance between consecutive waypoints needs to be kept suf?ciently small to retain high approximation accuracy. To resolve this fundamental issue, we propose in this paper a new and general framework for UA V trajectory and communication co-design with ?exible number of waypoint optimization variables (called designable waypoints) or their sub-path representations. First, we propose a ?exible path discretization scheme that optimizes only a number of selected waypoints (designable waypoints) along the UA V path for complexity reduction, while all the designable and non-designable waypoints ...

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1 Evolutionary Biparty Multiobjective UA V Path Planning: Problems and Empirical Comparisons Kesheng Chen, Wenjian Luo, Senior Member, IEEE, Xin Lin, Zhen Song, Yatong Chang Abstract?Unmanned aerial vehicles (UA Vs) have been widely used in urban missions, and proper planning of UA V paths can improve mission e?iciency while reducing the risk of potential third-party impact. Existing work has considered all e?iciency and safety objectives for a single decision-maker (DM) and regarded this as a multiobjective optimization problem (MOP). However, there is usually not a single DM but two DMs, i.e...

File: ssrn-6569619.pdf

Urban Air Mobility (UAM) is emerging as a groundbreaking solution to the growing challenges of urban transportation. It leverages machine learning (ML) technologies to improve efficiency, safety, and sustainability. This literature review paper explores the latest advancements in ML applications within the UAM domain, focusing on three key application areas: demand prediction for air taxi services, autonomous navigation in complex urban environments, and communication systems for UAM vehicles. In the contex t of air taxi services, a major use case of UAM applications, the ML algorithms are essential for predicting passenger demand, which helps optimize resource allocation and operational strategies. Accurate demand forecasting ensures that UAM services can effectively meet fluctuating urban transportation needs. For autonomous navigation, ML ?specifically, deep reinforcement learning ?enables UAM vehicles to navigate urban landscapes efficiently by optimizing flight trajectories. This minimizes energy consumptio n and reduces noise pollution, enhancing the overall sustainability of UAM operations. Moreover, ML is critical in improving UAM communication systems by addressing challen...

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Politecnico di Torino Master of Science degree in Mechatronic Engineering Master Thesis PX4 autopilot customization for non-standard gimbal and UWB peripherals Supervisor: Marcello Chiaberge Thesis advisors : Dott. Ing. Gianluca Dara Dott. Ing. Simone Silvestro Candidate: Francesco Malacarne s260199 Academic year 2019-2020

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displacement vector, we express the RP movement in polar form as wr[n] = wr[n? 1] + vr[n] [cos ?r[n] sin ?r[n]] , (1) where vr[n]? 0 and ?r[n]? [0, 2?) denote the step length and moving direction of the RP at slot n, respectively. The position of user k in the group is modeled as a bounded perturbation around the RP as wk[n] = wr[n] + rk[n] [cos ?k[n] sin ?k[n]] , ?k, (2) where rk[n]? [0, rmax d] denotes the deviation distance and ?k[n]? [0, 2?) denotes the deviation direction. For individual users, the position of an individual user k evolves as wk[n] = wk[n? 1] + vk[n] [cos ?k[n] sin ?k[n]] , ?k, (3) where vk[n]? 0 and ?k[n]? [0, 2?) denote the step length and moving direction. An individual user may join a nearby group with probability pjoin, provided that at least one reference point is within an attachment radius dg as shown in Fig. 2. Specifically, define the candidate set as Gk[n] ?{ g|? wk[n]? wg[n]?? dg} . (4) IfGk[n]?=?, then with probability pjoin user k switches to group mode and selects a group with the nearest reference point, Conversely, a group member may leave the group with probability pleave if it drifts beyond a detachment radius dg. At each time slot, both ...

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UA V Trajectory Optimization via Improved Noisy Deep Q-Network 1st Zhang Hengyu Dept. of Computer Science Wenzhou-Kean University Wenzhou, China zhanheng@kean.edu 2nd Maryam Cheraghy* Dept. of Computer Science Wenzhou-Kean University Wenzhou, China mcheragh@kean.edu *Corresponding author 3rd Liu Wei Dept. of Computer Science Wenzhou-Kean University Wenzhou, China lwei@kean.edu 4th Armin Farhadi School of Electrical and Computer Engineering University of Tehran Tehran, Iran armin.farhadi@ut.ac.ir 5th Meysam Soltanpour School of Computer Science and Info. Eng. Qilu Institute of Technology Jinan,...